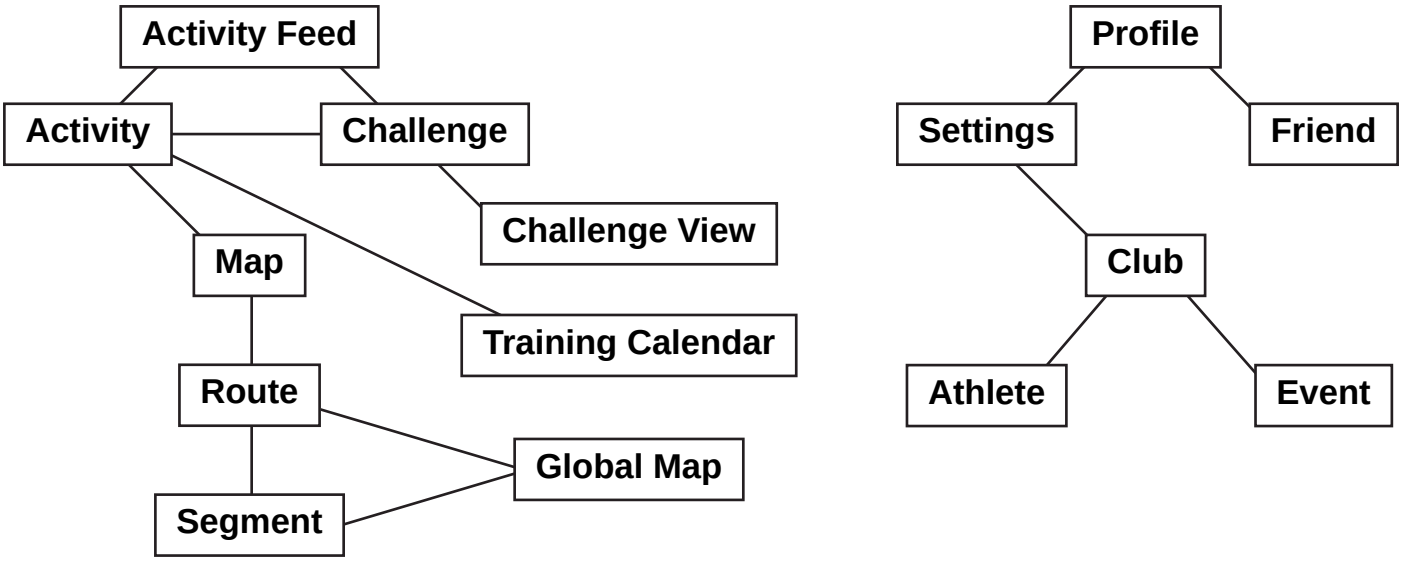
Project Activity: Time to Apply Object-Oriented Thinking!

Sample Solution: Strava.com



Conceptual model of Strava.com in October 2024.
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I created the conceptual model above with a UML modeling tool called Astah, but for this activity almost anything that allows rectangles and lines would do, including paper and pencil. The empty compartments in each rectangle are for the attributes and operations we would add later in the design process. Most UML tools allow you to hide these if you're not using them, but I've left them exposed in this instance.

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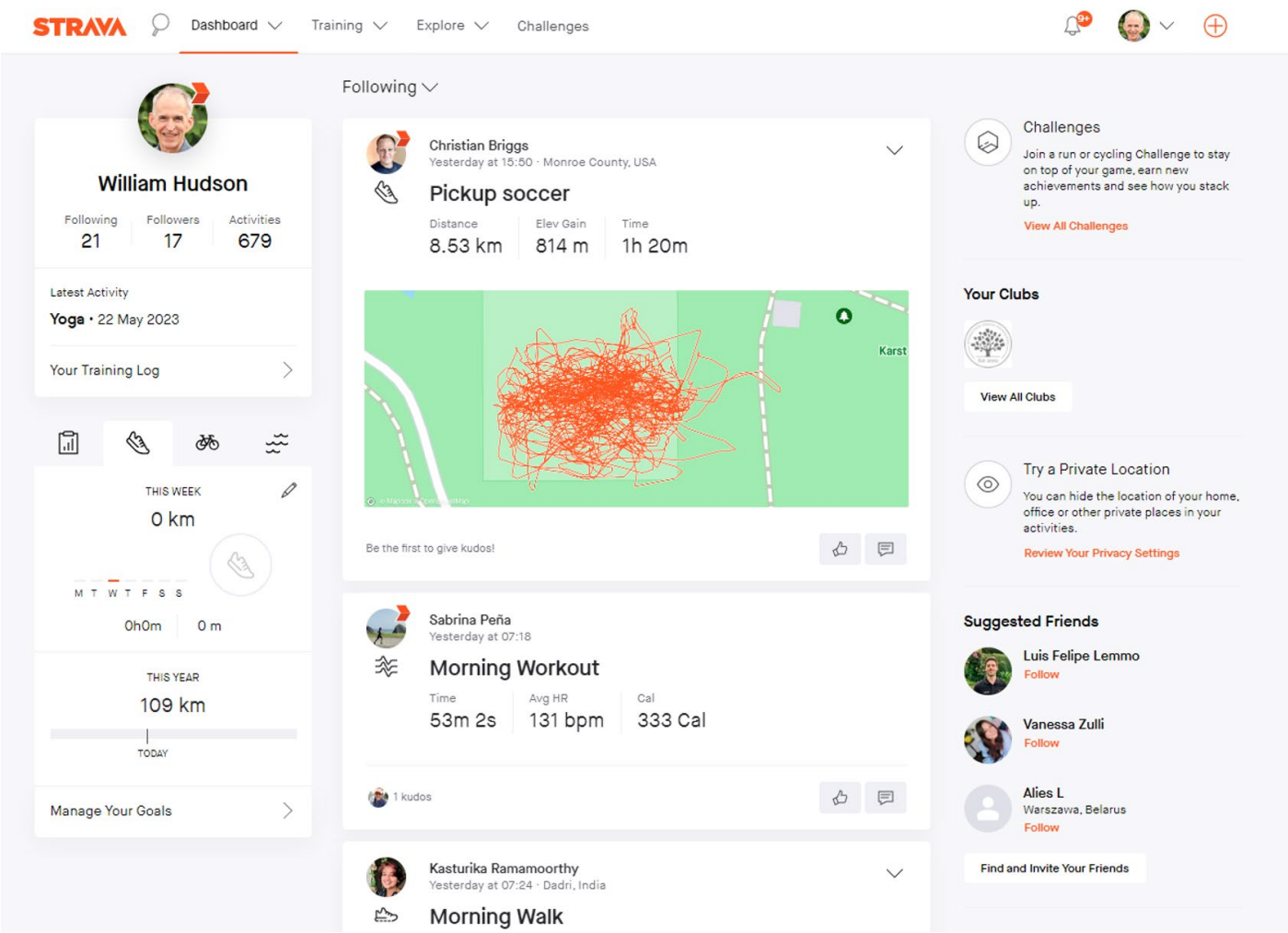

The Strava Dashboard, which features many of the concepts listed below.
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Below are the concepts I've included in the model:

Activity Feed: This is the main content of the homepage but isn't labeled. It includes your and your friends' activities along with news about club events, challenges, and Strava itself.

Activity: These are your logged activities, ranging from running to windsurfing. Each activity has numerous attributes according to type. For example, running and walking activities include a GPS map, heart rate (if monitored), cadence, distance, shoes used, and more.

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STRAVA Dashboard Training Explore Challenges

My Activities

Keywords

My Morning Workout | Search

Sport

All Sport Types

☐ Commute ☐ Private

679 Activities

Sport	Date	Title	Time	Distance	Elevation	Relative Effort			
Yoga	Mon, 22/05/2023	Yoga	1:00:00	0 km	0 m		Edit	Delete	Share
Hike	Sun, 21/05/2023	Abingdon to Sutton Courtenay Hike	2:44:34	10.77 km	18 m	25	Edit	Delete	Share
Yoga	Mon, 15/05/2023	Yoga	1:00:00	0 km	0 m		Edit	Delete	Share
Hike	Sun, 30/04/2023	Bluebell Hike	2:19:06	10.28 km	199 m		Edit	Delete	Share
Run	Fri, 28/04/2023	Morning Run	17:05	2.65 km	9 m	17	Edit	Delete	Share
Run	Wed, 26/04/2023	Morning Run	16:55	2.62 km	9 m	19	Edit	Delete	Share
Yoga	Mon, 24/04/2023	Yoga	1:00:00	0 km	0 m		Edit	Delete	Share

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Challenge: These are collective activities that apply over time. So, if I sign up for a challenge to walk 20 km in October, each walking activity in that month contributes to it. Finishing a challenge results in a virtual trophy, which can go in your virtual trophy cabinet (not included in the model above).

Map: Each GPS-tracked activity has an associated map that shows the route and segments.

Route: Saved routes that are available for others to use.

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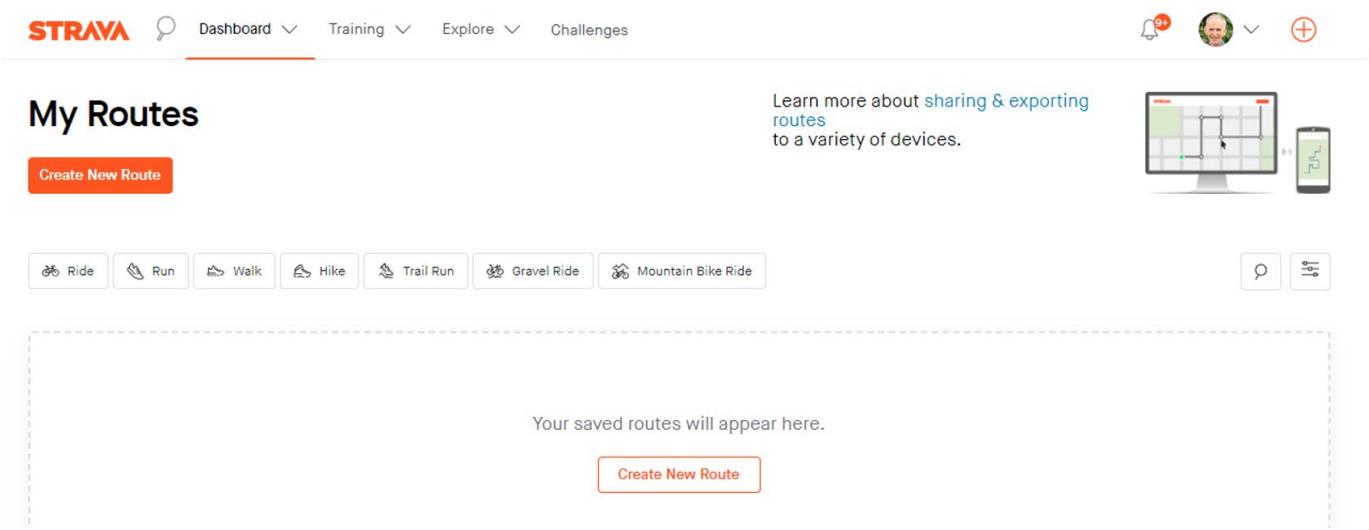

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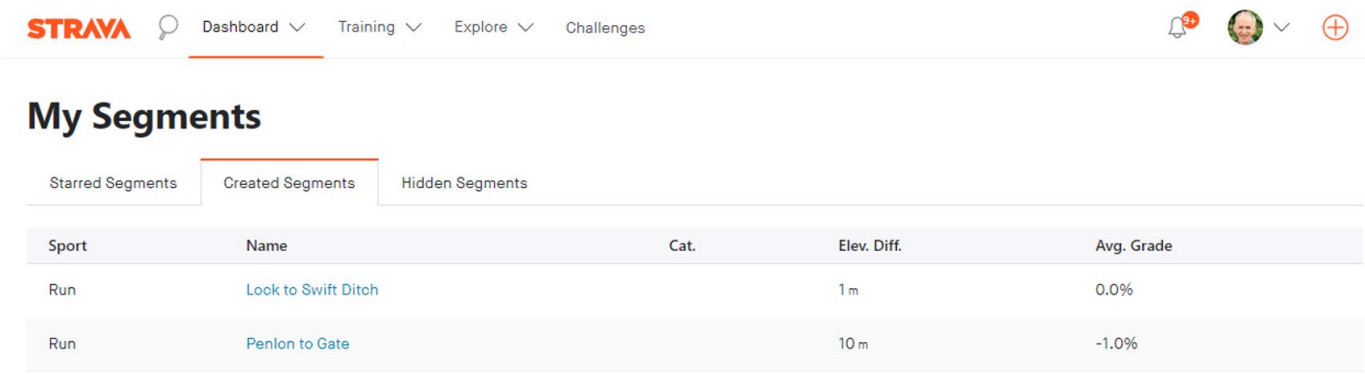
Segments: Defined paths with start and end points over which performance statistics and league tables are maintained.



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Challenge View: This is a separate view showing all available challenges. On the website, it is one of the pages in the main navigation. In the Android app, it appears under a separate tab called Groups. This creates some confusion when moving between the website and app.

Training Calendar: This is a monthly view of activities. It appears only on the website, not the app.

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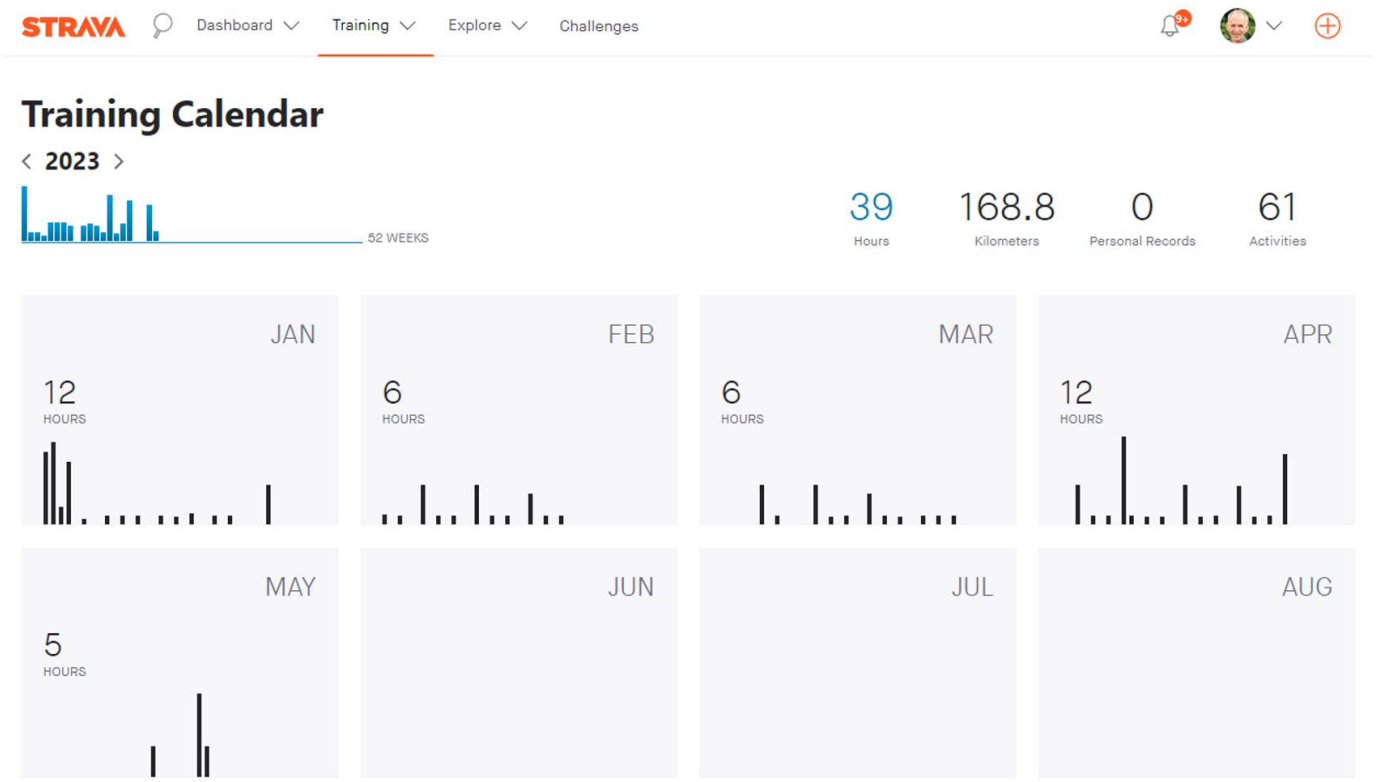

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Global Map: A summary map showing available routes and your choice of public or personal heatmaps of activities. (Routes are a subscription-only feature.)

Profile: A summary of your activities, trophies, stats, and more.

Settings: Personal details and many options that can be adjusted.

Friend: Another Strava member you are connected to. (In UML, names are singular by convention, but it doesn't really matter for our purposes.)

Club, Athlete, and Event: A club has athletes as members and can create events.

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